

MOTIVATION LETTER

Gawtam Chithra Ramesh

Computer Science

PhD Applicant

KTH Royal Institute of Technology, Stockholm

I want to build robot learning and control systems that are both data-efficient and provably safe, and I believe the right way to get there is through geometric structure and physics-based priors rather than scaling data collection. The graduate research position within the SAInt project, supervised by Asst. Prof. Noémie Jaquier, is where I see these ideas mattering most: developing motor intelligence for humanoid robots that physically assist older adults and people with special needs. Safe, compliant interaction with vulnerable users is not a setting where you can tolerate black-box policies that fail unpredictably. It demands the kind of principled, geometry-aware learning that the GeoRob Lab is built around. I am writing to apply for this position because it sits at the exact intersection of what I care about technically and what I find meaningful as an application.

Before KTH, I completed a dual degree at the Indian Institute of Technology Madras (B.Tech in Engineering Design and M.Tech in Robotics). That is where I first got serious about robotics. My dual degree project involved building a 3D perception system using structure-from-motion to reconstruct human torso geometry from monocular RGB images, then using ROS and MoveIt to plan ultrasound probe trajectories on the reconstructed surface. The project was not about geometry in the Riemannian sense, but it gave me an early appreciation for how geometric representations of physical structure can directly inform manipulation. I was awarded the IIT Madras Young Research Fellowship during this period.

I also spent eight months as a robotics intern at Systemantics, a startup building modular collaborative manipulators. I was involved in the full development cycle of a 6-DOF robot arm, from low-level firmware through trajectory generation and control in C++, to behavior tree-based task planning in ROS. Working with real hardware at that stage taught me things simulation cannot: the gap between a working algorithm and a working robot is often about the assumptions your model makes about the physical world. That lesson has stayed with me.

Since joining KTH, I have focused on learning-based approaches for robotic manipulation. Through a year-long research project at RPL under Prof. Florian Pokorny, I worked on deformable object manipulation. The core contribution was a taxonomy-guided framework that maps high-level task descriptions into structured motion primitive representations that a robot can execute. Rather than relying on end-to-end learned policies, the system uses structured intermediate representations to make the mapping from language to action interpretable and modular. I submitted this work to the ROMADO workshop at IROS 2025. The project reinforced something I kept running into: the gap between what large pretrained models can understand semantically and what a robot actually needs to execute reliably is not closing as fast as people assume. Structured representations, whether taxonomies, primitives, or geometric objects, still do essential work in bridging that gap.

In parallel, I spent six months as a research intern at Ericsson Research in Lund, developing robust 3D scene graph representations for real-world environments. I integrated instance segmentation, tracking, and RGB-D inpainting into Hydra, a state-of-the-art scene graph framework, to enable privacy-aware reconstruction and reduce sensitivity to parameter tuning. I also systematically analyzed multiple scene graph frameworks to identify open problems in object merging, robustness, and real-time deployment. This resulted in a first-author paper accepted at IMPROVE 2026 and a submission under review at ICPR 2026. The Ericsson experience was valuable for learning how to scope and execute research in a collaborative industrial setting and how to navigate peer review.

My current master's thesis at ABB Corporate Research is the work I consider most relevant to this position, and it is where my thinking about geometric structure in learned control has become most concrete. The problem is: given a manipulation task described by a Signal Temporal Logic (STL) specification, how do you generate safe control policies that satisfy temporal constraints while retaining the flexibility of learned, continuous-action models? My approach converts STL specifications into time-varying Control Barrier

Functions (CBFs), which provide continuous safety constraints that compose with the learning objective. In early experiments I built a hierarchical RL framework on top of this, using event-based STL decomposition and learning-from-demonstration to improve training efficiency. This works on simple 2D tasks so far.

Where I find the work most interesting, and where I see a clear line to the GeoRob Lab’s research, is in how I am thinking about the latent structure of the policy. Rather than treating logical constraints as flat text or static embeddings fed to a network, I represent STL predicates as structured regions in the policy’s latent space and interpret logical operators (conjunction, disjunction, temporal until) as geometric transformations over those regions. A flow-matching model then generates trajectories through this geometrically structured latent space. The bet is that if the latent geometry reflects the compositional structure of the task, you get better generalization to new specifications without retraining. The idea is still formative, but I think it points toward something real: that the right geometric structure in a learned representation is not just a convenience for data efficiency, it is what makes compositional generalization possible at all. Extending this to Riemannian formulations, where the manifold structure itself encodes constraints like joint limits, collision avoidance, or compliant interaction, feels like a natural and exciting next step.

This connects to what I want to pursue during the PhD. On the application side, I am drawn to contact-rich assistive manipulation, the kind of tasks SAInt targets, where you simply cannot collect millions of demonstrations and where safety during physical human-robot interaction is non-negotiable. Geometric priors and formal guarantees are not optional luxuries in that setting; they are requirements. On the theoretical side, I want to understand what manifold structure actually buys you in terms of sample complexity and out-of-distribution robustness. The question that the upcoming workshop on “Geometry in the Age of Data-Driven Robotics” poses, captures very interesting take that I would want to spend a PhD on, ‘is geometric structure complementary to data-driven learning, or is it the thing that makes data-driven learning work in the first place?’

Through my coursework at IIT Madras and KTH, I believe I have built relevant technical foundations to required to start exploring the project. I am proficient in Python and C++, experienced with PyTorch, JAX, ROS2, and MuJoCo. Having spent a year within the RPL community, I love the research culture and I know it is would be a great place to pursue my doctoral work.

Thank you for considering my application.